

NOT-SO-OPTIMAL TRANSPORT FLOWS FOR 3D POINT CLOUD GENERATION

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<https://research.nvidia.com/labs/genair/not-so-ot-flow>

ABSTRACT

Learning generative models of 3D point clouds is one of the fundamental problems in 3D generative learning. One of the key properties of point clouds is their permutation invariance, i.e., changing the order of points in a point cloud does not change the shape they represent. In this paper, we analyze the recently proposed equivariant OT flows that learn permutation invariant generative models for point-based molecular data and we show that these models scale poorly on large point clouds. Also, we observe learning (equivariant) OT flows is generally challenging since straightening flow trajectories makes the learned flow model complex at the beginning of the trajectory. To remedy these, we propose *not-so-optimal transport flow models* that obtain an approximate OT by an offline OT precomputation, enabling an efficient construction of OT pairs for training. During training, we can additionally construct a hybrid coupling by combining our approximate OT and independent coupling to make the target flow models easier to learn. In an extensive empirical study, we show that our proposed model outperforms prior diffusion- and flow-based approaches on a wide range of unconditional generation and shape completion on the ShapeNet benchmark.

1 INTRODUCTION

Generating 3D point clouds is one of the fundamental problems in 3D modeling with applications in shape generation, 3D reconstruction, 3D design, and perception for robotics and autonomous systems. Recently, diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020) and flow matching (Lipman et al., 2022) have become the de facto frameworks for learning generative models for 3D point clouds. These frameworks often overlook 3D point cloud permutation invariance, implying the rearrangement of points does not change the shape that they represent.

In closely related areas, equivariant optimal transport (OT) flows (Klein et al., 2024; Song et al., 2024) have been recently developed for 3D molecules that can be considered as sets of 3D atom coordinates. These frameworks learn permutation invariant generative models, i.e., all permutations of the set have the same likelihood under the learned generative distribution. They are trained using optimal transport between data and noise samples, yielding several key advantages including low sampling trajectory curvatures, low-variance training objectives, and fast sample generation (Pooladian et al., 2023). Albeit these theoretical advantages, our examination of these techniques for 3D point cloud generation reveals that they scale poorly for point cloud generation. This is mainly due to the fact that point clouds in practice consist of thousands of points whereas molecules are assumed to have tens of atoms in previous studies. Solving the sample-level OT mapping between a batch of training point clouds and noise samples is computationally expensive. Conversely, ignoring permutation invariance when solving batch-level OT (Pooladian et al., 2023; Tong et al., 2023) fails to produce high-quality OT due to the excessive possible permutations of point clouds.

In this paper, we propose a simple and scalable generative model for 3D point cloud generation using flow matching, coined as *not-so-optimal transport flow matching*, as shown in Fig 1. We first propose an efficient way to obtain an approximate OT between point cloud and noise samples. Instead of searching for an optimal permutation between point cloud and noise samples online during training, which is computationally expensive, we show that we can precompute an OT between a dense point superset and a dense noise superset offline. Since subsampling a superset preserves the

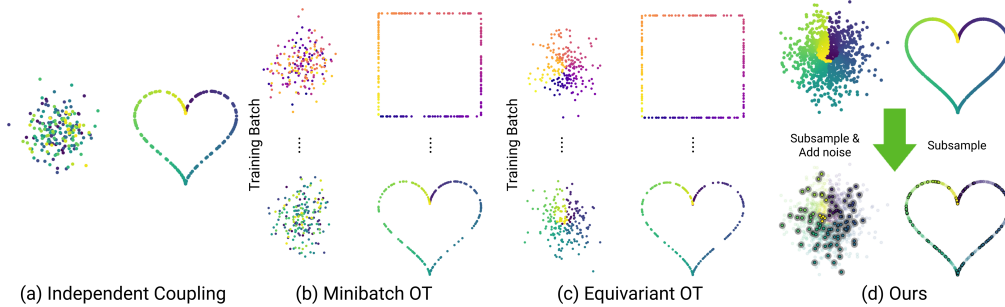


Figure 1: Different coupling types between Gaussian noise (left) and point clouds (right), where coupled noise and surface points share the same color: (a) Independent Coupling randomly maps noises to point clouds. (b) Minibatch OT computes OT map in batches of noises and point clouds. (c) Equivariant OT follows the similar minibatch OT but aligns points via permutation. (d) Our approach precomputes dense OT on data and noise supersets, then subsamples it to couple point clouds with slightly perturbed noise. Note that only (c) and (d) can produce high-quality OT.

underlying shape, we can simply subsample the point superset and obtain corresponding noise from the precomputed OT to construct a batch of noise-data pairs for training the flow models.

We demonstrate that our approximate OT reduces the pairwise distance between data and noise significantly and benefits from the advantages of OT flows, *e.g.*, straightness of trajectories and fast sampling. However, a careful examination shows that learning (equivariant) OT flows is generally challenging since straightening flow trajectories makes the learned flows complex at the beginning of the trajectory. Intuitively, in the OT coupling, the flow model should be able to switch between different target point clouds (i.e., different modes in the data distribution) with small variations in their input, making the flow model have high Lipchitz (see Fig 2).

To remedy this, we propose a simple approach to construct a less “optimal” hybrid coupling by blending our approximate OT and independent coupling used in the flow matching model. In particular, we suggest perturbing the noise samples obtained from our approximate OT with small Gaussian noise. While this remedy makes our mapping less optimal from the OT perspective, we show that it empirically shows two main advantages: First, the target flow model is less complex and the generated points clouds have high sample quality. Second, when reducing the number of inference steps, the generation quality still degrades slower than other competing techniques, indicating smoother trajectories.

In summary, this paper makes the following contributions: (i) We show that existing OT approximations either scale poorly or produce low-quality OT for real-world point cloud generation. (ii) We show that albeit the nice theoretical advantages, equivariant OT flows have to learn a complex function with high Lipchitz at the beginning of the generation process. (iii) To tackle these issues, we propose a not-so-optimal transport flow matching approach that involves an offline superset OT precomputation and online random subsampling to obtain an approximate OT, and adds a small perturbation to the obtained noise during training. (iv) We empirically compare our method against diffusion models, flows, and OT flows on unconditional point cloud generation and shape completion on the ShapeNet benchmark. We show that our proposed model outperforms these frameworks for different sampling budgets over various competing baselines on the unconditional generative task. In addition, we show that we can obtain reasonable generation quality on the shape completion task in less than five steps, which is challenging for other approaches.

2 RELATED WORKS

Score-based Generative Models. Recently, diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song & Ermon, 2019) have gained popularity in generating data of various formats, especially images (Rombach et al., 2022; Ramesh et al., 2022) and videos (Blattmann et al., 2023; Brooks et al., 2024). These generative models employ an iterative process to transform a known distribution,

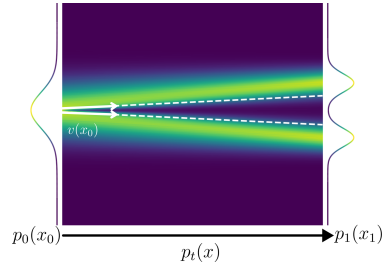


Figure 2: In the OT flow model, the vector field $\mathbf{v}_t(\mathbf{x}_0)$ admits a large change in its output with a small perturbation of $\mathbf{x}_0$ at $t=0$.

typically a Gaussian, into a desired data distribution. Song et al. (2020) demonstrate that these models can be generalized to a continuous time setting, solving an SDE with an ODE that shares the same marginal. Lipman et al. (2022) train a vector field to trace a linear interpolation between training pairs from data and noise distribution (further details in Section 3.1). We also focus on the flow matching models and aim to employ them effectively for 3D point cloud generation.

Relation of Flow Matching with Optimal Transport (OT). Flow matching is closely related to the Optimal Transport (OT) problem, which aims to find a map with minimal transport cost between two distributions. As for flow matching, the trajectory flows can be taken to define the map that solves the OT problem. However, when the flow models are trained with random pairs, the trajectories are usually curved, so using pairs from the OT map can lead to straighter trajectories that improve the efficiency, as Pooladian et al. (2023); Tong et al. (2023) show. Since the OT map is usually unavailable, various methods are introduced to obtain better training pairs as we discuss in Section 3.1. Another approach to obtain better pairs is to iteratively straighten the trajectory from a flow matching model. Rectified Flow (Liu et al., 2022) obtains training pairs by simulating ODE trajectories of pre-trained models. However, this involves an expensive ODE simulation and introduces errors in the straightening process. In this work, we mainly study the limitations of existing OT solutions on the first approach and derive a new solution that is specific for point cloud generation.

3D Point Cloud Generation. Point cloud generation has been studied extensively using various generative models including VAEs (Kim et al., 2021), Energy-based models (Xie et al., 2021), GANs (Achlioptas et al., 2018; Shu et al., 2019; Li et al., 2021), and flow-based models (Yang et al., 2019; Kim et al., 2020). Recently, diffusion models (Zeng et al., 2022; Zhou et al., 2021; Luo & Hu, 2021; Peng et al., 2024; Zhao et al., 2025) have achieved great success in producing high-quality generation results. However, these methods often ignore point clouds’ permutation-invariant nature in training by treating them as high-dimensional flat vectors. PSF (Wu et al., 2023) uses Rectified flow (Liu et al., 2022) to generate point clouds, addressing permutation issues implicitly through flow straightening. This approach is computationally expensive due to multiple sample inferences needed for constructing training pairs. In this work, we focus on constructing an efficient, high-quality OT approximation for permutation-invariant point cloud generation based on the flow matching models.

3 METHOD

In Section 3.1, we begin by covering preliminaries of training a continuous normalizing flow and recent OT flows. Section 3.2 explores the challenges of applying existing OT approximation methods to 3D point clouds. To tackle these challenges, we introduce our approximate OT approach in Section 3.3 that precomputes OT maps in an offline fashion, and in Section 3.4, we explore a simple hybrid and less optimal coupling approach that makes target flows easier to learn.

3.1 PRELIMINARIES

Continuous Normalizing Flow (CNF) (Chen et al., 2018) morph a base Gaussian distribution q_0 into a data distribution q_1 using a time-variant vector field $\mathbf{v}_{\theta,t} : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$, parameterized by a neural network θ . The mapping is obtained from an ordinary differential equation (ODE):

$$\frac{d}{dt}\mathbf{x}_t = \mathbf{v}_{\theta,t}(\mathbf{x}_t). \quad (1)$$

Conceptually, the ODE transports an initial sample $\mathbf{x}_0 \sim q_0$, where $\mathbf{x}_0 \in \mathbb{R}^d$ with p_t denoting the distribution of samples at step t and $p_0(\mathbf{x}) := q_0(\mathbf{x})$. Usually, the vector field $\mathbf{v}_{\theta,t}$ is trained to maximize the likelihood p_1 assigned to training data samples $\mathbf{x}_1$ from distribution q_1 . This procedure is computationally expensive due to extensive ODE simulation for each parameter update.

Flow Matching (Lipman et al., 2022) avoid the computationally expensive simulation process for training CNFs. In particular, we define a conditional vector field $\mathbf{u}_t(\cdot|\mathbf{x}_1)$ and path $p_t(\cdot|\mathbf{x}_1)$ that transform q_0 into a Dirac delta at $\mathbf{x}_1$ at $t = 1$. Lipman et al. (2022) show that $\mathbf{v}_{\theta,t}$ can be learned via a simple conditional flow matching (CFM) objective:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t,q_1(\mathbf{x}_1),q_0(\mathbf{x}_0)} \|\mathbf{v}_{\theta,t}(\mathbf{x}_t) - \mathbf{u}_t(\mathbf{x}_t|\mathbf{x}_1)\|^2. \quad (2)$$

A common choice for the conditional vector field is $\mathbf{u}_t(\mathbf{x}|\mathbf{x}_1) := \mathbf{x}_1 - \mathbf{x}_0$, which can be easily simulated by linearly interpolating the data and Gaussian samples via $\mathbf{x}_t = (1 - t) * \mathbf{x}_0 + t\mathbf{x}_1$.